

Use Case “Add Product”

1. Use case code

UC234.1

2. Brief Description

This use case allows the Product Manager to add a new media product (Book, Newspaper, CD, DVD) to the AIMS system. The system will check data format constraints and pricing rules before officially saving it to the database.

3. Actors

3.1 Product Manager

4. Preconditions

- Product Manager successfully logged into the AIMS System with appropriate account provided.
- The AIMS System is currently operating normally..

5. Basic Flow of Events

1. Product Manager request access to the product management interface.
2. Software displays list of products and options for adding, editing, deleting and adjusting quantities.
3. Product Manager select Add Product
4. Software displays general form for Product Manager
5. Product Manager input general information and choose specific type of product.
6. Software displays form for fill specific information depending on product type
7. Product Manager input specific information and submit.
8. Software checks the data and price rule.
9. Software update product list
10. Software display notifications and save record into history

6. Alternative flows

Table 1 - Alternative flows of events for UC “Add Product”

No	Location	Condition	Action	Resume location
1.	At step 8	If the input data is not in the correct format	▪ Software display detailed error for each data field	At step 3 or 5
2.	Any step	If Product Manager cancel process	▪ Software do not save any changes and return product list interface	End the use case

7. Input data

Table 2 - Input data of UC “Add Product”

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Barcode	Unique identifier for the product	Yes	Unique in the system; numeric only	893123456789
2.	Title	The name of the media product	Yes	Cannot be blank	DVD Hello World
3.	Category	Product classification	Yes	One of: Books, Newspapers, CDs, DVDs	Books
4.	General Description	Condition, primary color, or return condition	Yes	Text description	New condition, blue cover
5.	Original Value	Base price excluding 10% VAT	Yes	Positive integer	100
6.	Current Price	Selling price excluding 10% VAT	Yes	Between 30% and 150% of Original Value	120
7.	Dimensions	Physical size (Height x Width x Length)	Yes	Positive values only	20 x 15 x 3 (cm)
8.	Weight	Physical weight of the product	Yes	Positive value (kg)	0.5
9.	Author	Creator(s) of the book	Yes (Book)	Required if Category is "Books"	Ngo Tat To
10.	Cover Type	Format of the book cover	Yes (Book)	"Paperback" or "Hardcover"	Paperback
11.	Editor-in-chief	Head of the newspaper	Yes (News)	Required if Category is "Newspapers"	Nguyen Van A
12.	Artists	Performers or singers	Yes (CD)	Required if Category is "CDs"	Son Tung M-TP
13.	Track List	List of songs with titles and lengths	Yes (CD)	Must include title and duration per track	Track 1 - 10:30
14.	Disc Type	Physical format of the disc	Yes (DVD)	"Blu-ray" or "HD-DVD"	Blu-ray
15.	Director	Director of the media	Yes (DVD)	Required if Category is "DVDs"	Christopher Nolan

1. Output data

Table 3 - Output data of UC “Add Product”

No	Data fields	Description	Display format	Example
1.	Operation Status	Notification of the result for the performed action (add, edit, delete, or adjust).	Text notification	Product added successfully
2.	Error Message	Detailed notification when input data is invalid or violates system rules.	Text (Red)	Price must be between 30% and 150% of Original Value
3.	History Log	A record list of all product additions, edits, and deletions.	Timeline/Log entry	16/03/2026: Manager A added product XXXX
4.	Product List	A list of all media products currently available in the store's inventory.	Data table	(List of Books, CDs, ...)

2. Postconditions

- Interface Refresh: The product management interface is automatically refreshed to display the most recent data after the operation.
- Audit trail: A new record created in history, log the detail information of the operation
- User notification: Product Manager receive notification from software, product list interface must refreshed to display updated list
- Stored data: All products successfully saved into database or new information be updated with newest version.
- Error message: Product Manager receive error text message at invalid field.